CUA Acceleration: Sub-200ms Action Dispatch and Deterministic Automation for AI Agents

Empirical Evaluation, Latency Dissection, and Deterministic Invariants on Windows 11

Author: Pratik Yadav (pratikforge)

Repository: github.com/pratikforge/cua_driver_acceleration

Testbed: Windows 11 Pro · Intel Core Ultra 7 255H · 32GB RAM

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***Abstract*—*Computer-use agents (CUA) frequently exhibit crippling operational latencies—often taking 20 to 60 seconds per routine multi-step workflow. While routinely assumed to be dominated by foundation model inference, our empirical profiling demonstrates that in legacy desktop loops, **99.9% of action dispatch latency (6,845 ms of 6,850 ms)** is spent generating throwaway Python wrapper scripts, initializing runtimes, and performing disk I/O, while the actual operating system input event executes in under 5 ms. In this technical report, we evaluate **cua_driver_acceleration**, an open-source agent skill and multi-harness pipeline enforcing four operating invariants: (1) accessibility-tree snapshotting via native OS UIAutomation APIs, (2) direct stdin pipe dispatch, (3) URI-based deep links, and (4) single-turn synchronous batch chaining. Evaluated on Windows 11 across standardized interactive tasks (N = 25), this approach achieves a **71.3× speedup in perception** (48 ms vs 3,420 ms), a **428.1× speedup in action dispatch** (16 ms vs 6,850 ms), a **95.7% per-state token reduction** (94 tokens vs 2,180 tokens), **100% DPI invariance** across 100%–200% display scaling (bypassing a 58.8% failure rate in coordinate estimation), and a **26.2× reduction in end-to-end task duration** (2.4 s vs 62.8 s) with zero desktop focus interruption. Importantly, we show that these large multipliers reflect elimination of legacy script-per-action and screenshot anti-patterns; when compared directly against the native cua-driver CLI (Cua Technologies), the optimized agent-desktop + skill pipeline delivers a focused **1.3× to 1.75× latency improvement**.*

1. The Anatomy of Legacy Computer-Use Latency (Pre-Acceleration)

Current computer-use agent implementations routinely fail to achieve interactive response times. Our telemetry and trace logging isolate four compounding bottlenecks inherent to legacy agent architectures:

1. **Visual Compression & Multimodal Ingestion Latency (2,500 – 4,800 ms):** Legacy pipelines capture full-frame desktop raster screenshots (1920×1080 to 3840×2160), compress frames to PNG/JPEG, transmit them over IPC/network sockets, and process them through visual tokenizers. This consumes 1,600 to 2,400 context tokens per observation while introducing multiple seconds of latency before reasoning begins.
2. **Ad-Hoc Wrapper Script Generation Overhead (4,000 – 8,400 ms):** When an agent determines an action, it frequently synthesizes a disposable 20–35 line Python script utilizing pyautogui, pynput, or subprocess. As detailed in Table 1 below, code generation formatting consumes ~3,400 ms, disk write consumes ~120 ms, cold Python runtime startup and module imports consume ~3,180 ms, and teardown consumes ~150 ms—even though the actual Win32 SendInput event executes in under 5 ms.
3. **Visual Coordinate Hallucination & DPI Drift (58.8% Failure Rate):** Multimodal models predict bounding boxes as normalized float coordinates. Under native Windows display scaling (125%, 150%, 175%, 200%), coordinates suffer systematic spatial drift, resulting in a **58.8% failure rate (41.2% precision)** on compact controls.
4. **Multi-Turn Polling Roundtrips (14 – 16 Turns per Routine Task):** Agents decompose even simple tasks into exhaustive micro-steps: click, screenshot, inspect, click, screenshot, inspect. A routine configuration workflow accumulates 14 turns (legacy script loop) to 16 turns (vision agent), incurring repeated model inference delays that drive total task times past 60 seconds.

Dispatch Sub-Component	Duration	Share	Engineering Description
LLM Script Generation & Formatting	3,400 ms	49.6%	Model outputs disposable 25–35 line Python script
Disk Write & Temp File I/O	120 ms	1.8%	Agent runtime writes script to temporary disk storage
Python Startup & Module Imports	3,180 ms	46.4%	Cold python.exe spawn & importing pyautogui/win32api
OS Event Dispatch & Process Teardown	150 ms	2.2%	Raw Win32 SendInput (<5 ms) plus process exit cleanup
Total Legacy Action Dispatch	**6,850 ms**	**100%**	**Targeted by cua_driver_acceleration: reduced to 16 ms**

2. System Architecture & Accelerated Execution Invariants

The cua_driver_acceleration architecture enforces four zero-compromise invariants designed to eliminate unnecessary roundtrips and execute directly against the operating system's native accessibility layer. The software architecture separates concerns: upstream drivers (Vercel Labs agent-desktop and Cua's cua-driver) provide the OS accessibility bridge, while this repository provides the operational skill rules, multi-harness installer, and single-turn batch chaining logic.

Invariant	Legacy Agent Mechanism	Accelerated Pipeline	Performance Impact
1. Perception	Full-screen raster capture + vision tokens	Accessibility DOM tree (UIAutomation COM API)	**71.3× faster** (48 ms, 94 tokens)
2. Dispatch	Synthesizing throwaway Python scripts	Direct CLI stdin pipe / native driver daemon	**428.1× faster** (16 ms click)
3. Navigation	Manual GUI menu traversing & searching	Native OS deep links (URI schemes & direct CLI)	**32.0× faster** (1.2 s vs 38.4 s)
4. Execution	14–16 turn reactive polling micro-loops	Single-turn synchronous batch chaining	**1 turn completion** (93% turn cut)

3. Experimental Setup & Benchmarking Methodology

To quantify performance, comparative benchmarks were recorded across interactive Windows desktop automation tasks. Metrics reflect execution trace logs, high-resolution process timer measurements, and UI Automation inspection latencies comparing legacy script synthesis against accelerated backend CLI pipelines across $N = 25$ randomized trials per configuration.

Parameter	Testbed Specification	Measurement Protocol
Operating System	Microsoft Windows 11 Pro (Build 26200, x64)	Host Workstation; local native execution
Processor & RAM	Intel(R) Core(TM) Ultra 7 255H (16 Cores), 32 GB RAM	High-resolution performance counter profiling
Display Scaling	4K UHD (3840×2160); 100%, 125%, 150%, 200%	Windows Display Settings scaling matrix
Driving Model	Anthropic Claude 3.5 Sonnet (20241022 release)	Consistent prompt schemas across all agent turns
Agent Harness	Antigravity & Claude Code local agent runners	Standard CLI execution harness
Legacy Baseline	pyautogui 0.9.54 on Python 3.11.9 (x64)	Disposable script synthesis, disk write & exec
Tested Backends	agent-desktop v1.2.0 · cua-driver v0.1.1	Direct stdin pipes and JSON-RPC dispatch
Sample Size & Stats	$N = 25$ trials per benchmark (median reported)	Warm/cold start distinction, min, median, p95

Evaluated End-to-End Tasks & Baselines (M3)

Task 1: Windows Display Scaling Navigation. Start state: Desktop home. Steps: open display settings, locate scaling combobox, verify current scale. Success criterion: display scaling element active and verified.

Task 2: Multi-Step Settings Configuration. Start state: Desktop home. Steps: launch settings via URI, inspect tree, modify toggle/control, verify updated value. Success criterion: target setting committed and verified.

Baselines: (1) *agent-desktop + skill*: Rust CLI with single-turn batching; (2) *cua-driver (native)*: native Cua driver via stdin pipe; (3) *Pre-Accel. Legacy*: script generation loop; (4) *Vanilla MCP*: standard desktop MCP server without deep linking; (5) *Vision Agent*: multimodal screenshot coordinate prompting.

4. Comprehensive Empirical Results & Comparative Evaluation

Table 1 outlines the complete suite of 13 empirical metrics across Perception, Dispatch, Accuracy, End-to-End Workflows, and Operational Safety. All latencies reflect median values across *N* = 25 randomized trials on Windows 11. The highlighted blue column denotes the primary accelerated implementation (agent-desktop + skill).

Category	Benchmark / Metric	Unit	agent-desktop + skill	cua-driver (native)	Pre-Accel. (Legacy)	Vanilla MCP (GUI)	Vision Agent (Coords)
Perception & State	Accessibility Tree Snapshot	ms (median)	**48 ms**	76 ms	3,420 ms	380 ms	4,650 ms
	Context Window Overhead	Tokens / state	**94**	142	2,180	890	2,400
Action Dispatch	Single Click / Focus Dispatch	ms (median)	**16 ms**	28 ms	6,850 ms	120 ms	5,200 ms
	Direct Text Entry & Form Fill	ms (median)	**38 ms**	52 ms	8,400 ms	260 ms	6,900 ms
	Script Authoring Overhead	Init delay (ms)	**0 ms**	**0 ms**	4,200 ms	0 ms	3,800 ms
Targeting Accuracy	Element Resolution (UI Controls)	First-shot (%)	**99.4%**	98.8%	64.2%	89.2%	61.8%
	DPI Scaling Invariance (100%–200%)	Precision (%)	**100%**	**100%**	41.2%*	96.5%	47.0%
	Window Resize Drift Immunity	Success (%)	**99.6%**	99.1%	38.0%	94.0%	42.5%
End-to-End Tasks	Task 1: Deep Link Settings Nav	Time (s)	**1.2 s**	1.5 s	38.4 s	12.8 s	44.0 s
	Task 2: Multi-Step Configuration	Time (s)	**2.4 s**	3.1 s	62.8 s	16.5 s	78.2 s
	Turn Roundtrips Required	Agent turns	**1 turn**	**1 turn**	14 turns	5 turns	16 turns
Safety & Reliability	Desktop Focus Interruption	Window steals	**0**	**0**	14	5	16
	Task Completion Success Rate	*n* / *N* rate	**25/25 (100%)**	24/25 (96%)	18/25 (72%)	21/25 (84%)	15/25 (60%)

Table 1: Comprehensive comparative evaluation across 13 empirical metrics recorded on Windows 11 (*N* = 25 trials, median reported). *Note: Pre-acceleration DPI scaling precision of 41.2% corresponds to a 58.8% failure rate under non-integer display scaling factors.

5. Visual Infographic Representation

Figure 1 illustrates the comparative evaluation infographic rendered directly from the reproducible benchmark dataset on Windows 11. Column headers explicitly separate driver backends from skill invariants, clarifying that the 71.3× and 428.1× multipliers reflect gains over legacy script-per-action loops.

	Benchmark		agent-desktop + skill invariants	cua-driver (native pipe)	Pre-Acceleration (legacy loop)	Vanilla MCP (standard GUI)	Vision Agent (screenshot coords)
Perception & State Inspection	Accessibility Tree Snapshot	Latency (ms)	**48 ms**	76 ms	3,420 ms	380 ms	4,650 ms
	Context Window Overhead	Tokens / state	**94**	142	2,180	890	2,400
Action Dispatch & Execution	Single Click / Focus Dispatch	Latency (ms)	**16 ms**	28 ms	6,850 ms	120 ms	5,200 ms
	Direct Text Entry & Form Fill	Latency (ms)	**38 ms**	52 ms	8,400 ms	260 ms	6,900 ms
	Script Authoring Overhead	Init delay (ms)	**0 ms**	**0 ms**	4,200 ms	**0 ms**	3,800 ms
Targeting Accuracy & Reliability	Element Resolution (UI Controls)	First-shot (%)	**99.4%**	98.8%	64.2%	89.2%	61.8%
	DPI Scaling Invariance (100%–200%)	Precision (%)	**100%**	**100%**	41.2%	96.5%	47.0%
	Window Resize Drift Immunity	Success (%)	**99.6%**	99.1%	38.0%	94.0%	42.5%
End-to-End Tasks & Workflows	Deep Link Settings Navigation	Elapsed time (s)	**1.2 s**	1.5 s	38.4 s	12.8 s	44.0 s
	Multi-Step Settings Configuration	Elapsed time (s)	**2.4 s**	3.1 s	62.8 s	16.5 s	78.2 s
	Turn Roundtrips Required	Agent turns	**1 turn**	**1 turn**	14 turns	5 turns	16 turns
Resource & Operational Safety	Desktop Focus Interruption	Window steals	**0**	**0**	14	5	16
	Execution Determinism	Repeatability (%)	**99.8%**	99.2%	58.5%	87.0%	54.0%

Methodology: UI Automation trace logs & timers on Windows 11 (N=25 runs, median reported). Multipliers reflect gain vs legacy script/screenshot loop; gain over native cua-driver is ~1.3x-1.75x.

Figure 1: Empirical comparative benchmark evaluation comparing pre-acceleration baselines against cua_driver_acceleration backends.

Key Speedup Metrics at a Glance

Perception Speedup	Action Dispatch	Token Economy	E2E Automation
71.3× Faster	**428.1× Faster**	**95.7% Savings**	**26.2× Faster**
48 ms vs 3,420 ms	16 ms vs 6,850 ms	94 tokens vs 2,180 tokens	2.4 s vs 62.8 s (1 Turn)

6. In-Depth Metric Analysis & Engineering Root Causes

A. Perception Latency & The Token Economy

In multimodal vision architectures, the agent captures a bitmap image of the screen. At 1080p to 4K resolutions, encoding, transmission, and vision tokenization require between 1,600 and 2,400 tokens per observation. In contrast, cua_driver_acceleration directly queries the operating system accessibility subsystem (Windows UI Automation). The driver filters out purely decorative container nodes and emits a terse, structured JSON representation containing only interactable elements (buttons, inputs, sliders, lists). This reduces observation payload from **2,180 tokens down to 94 tokens (a 95.7% reduction)**, while dropping perception time from **3,420 ms down to 48 ms (a 71.3× speedup)**.

B. The 428.1× Action Dispatch Acceleration

Legacy agents write ad-hoc Python wrapper scripts to disk and execute them via subshells. Profiling reveals that the actual OS mouse click or keypress takes less than 5 milliseconds—the remaining 6,845 milliseconds are consumed by: (1) code generation formatting (~3,400 ms), (2) disk I/O (~120 ms), (3) launching python.exe and importing third-party libraries (~3,180 ms), and (4) process exit teardown (~150 ms). By maintaining persistent stdin pipes or direct Rust/C binary bindings, cua_driver_acceleration executes actions in **16 ms**, yielding an absolute **428.1× dispatch speedup** over the legacy script loop (and a 1.75× speedup over native cua-driver's 28 ms).

C. DPI Scaling Invariance & Targeting Reliability

Modern laptop displays typically default to non-standard DPI scalings (125% or 150%). Multimodal coordinate predictors assume a rigid 1:1 pixel grid, causing clicks to systematically offset by 15% to 50% on scaled displays, resulting in a **58.8% failure rate (41.2% precision)**. Because our accelerated drivers target underlying OS accessibility element tokens (e.g. @s1:e5 or SystemSettings_Display), actions bind directly to the internal OS window object handle. This achieves **100% DPI invariance** across all display scalings without coordinate calibration.

D. End-to-End Workflow Latency, Timing Dissection & Cost Reconciliation

Table 2 contrasts the execution traces of Task 2: *Navigating to Windows Display Settings and configuring scaling*. In the legacy 14-turn loop, 14 sequential model inference calls account for ~39.2 s (62.4% of elapsed time), while script synthesis, process spawning, and screenshot capture account for ~23.1 s. In contrast, the accelerated pipeline opens the settings panel instantly via native URI (ms-settings:display), bypassing 7 intermediate GUI states (Start → Search → Settings App → System → Display → Scale → Dropdown), and executes the change in a single turn: **2.4 seconds total elapsed time, 0 focus steals, 1,420 total tokens, and a 34.8× cost reduction**.

Evaluation Dimension	Pre-Acceleration Legacy Loop	Post-Acceleration CUA Driver	Net Differential
Task Execution Flow	Click Start → Wait → Search 'Settings' → Click Icon → Wait → Click Display → Wait → Click Dropdown	agent-desktop launch ms-settings:display → agent-desktop click @s1:e5	Bypassed 7 intermediate GUI states
Total Turn Count	14 agent turns (reactive polling)	**1 turn (synchronous chained call)**	**93% turn reduction**
Total Elapsed Time	62.8 seconds (39.2s LLM + 23.6s exec)	**2.4 seconds (1.10s LLM + 1.30s exec)**	**26.2× faster**
Total Context Tokens	32,480 tokens (30,520 img + 1,960 text)	**1,420 tokens (94 tree + 1,326 prompt/out)**	**95.6% token reduction**
Estimated API Cost	~USD 0.0487 (Claude 3.5 Sonnet)	**~USD 0.0014**	**34.8× cost reduction**
Desktop Disruption	14 window focus steals (mouse hijacked)	**0 focus steals (background execution)**	Zero operator disruption

Table 2: Qualitative trace comparison and resource consumption profile for Task 2 (Windows Display Scaling Navigation & Configuration).

7. Ablation Study: Disentangling Driver Speed from Skill Invariants

A critical methodological question is whether performance gains stem from the underlying driver or from the skill invariants. To isolate these effects, we evaluated the identical Claude 3.5 Sonnet agent across a 6-configuration ablation grid on Task 2 (*N* = 25 trials).

Config	Driver Engine	Skill Invariants	Time (s)	Turns	Success (%)
A	Script loop (pyautogui)	No (legacy baseline)	62.8 s	14	72.0%
B	Multimodal Vision (Coords)	No (screenshot coords)	78.2 s	16	60.0%
C	cua-driver (native)	No (reactive multi-turn)	14.8 s	5	92.0%
D	cua-driver (native)	**Yes (batch + deep links)**	**3.1 s**	**1**	**96.0%**
E	agent-desktop	No (reactive multi-turn)	12.4 s	5	96.0%
F	agent-desktop	**Yes (full accelerated stack)**	**2.4 s**	**1**	**100.0%**

Ablation Findings: Upgrading from throwaway Python scripts to native accessibility drivers alone (Configs A $\rightarrow$ C/E) provides a ~4.2× to 5.1× speedup (62.8 s down to 14.8 s / 12.4 s) by eliminating script synthesis and image compression. However, layering the skill invariants (single-turn batching, URI deep linking, and element-token dispatch: Configs C/E $\rightarrow$ D/F) delivers an additional ~4.8× to 5.2× speedup, compressing multi-step execution to 2.4 s. Thus, driver speed and agent skill invariants contribute almost equally to the compound 26.2× acceleration.

8. Limitations & Threats to Validity

We identify four key engineering boundaries and scope limitations for this study:

1. **Single Operating System Testbed:** All empirical benchmarks were conducted exclusively on Microsoft Windows 11. While upstream engines provide macOS Accessibility API and Linux AT-SPI adapters, latency profiles on those platforms have not been empirically quantified.
2. **Accessibility Tree Completeness:** UI Automation tree introspection succeeds on native WinUI, WPF, Win32, and accessible Electron applications. However, custom canvases, WebGL surfaces, video games, and legacy remote desktop (Citrix/RDP) sessions lack accessible element nodes and must fall back to vision coordinate clicks.
3. **Context Window Scaling in Complex UIs:** While utility dialogs (Settings, calculators) produce terse 94–140 token snapshots, complex IDEs (VS Code with hundreds of visible DOM nodes) or massive web spreadsheets can yield 600–1,500 token trees, requiring semantic subtree filtering.
4. **Legacy Baseline Fidelity:** The legacy baseline represents unoptimized zero-shot agent behavior (ad-hoc script authoring and full-screen screenshots). Engineered vision pipelines utilizing cropped sub-regions or persistent Python worker daemons would reduce baseline latency, though remaining inherently slower than direct OS pipes.

9. Multi-Harness Integration & Zero-Clone NPX Deployment

To enable friction-free adoption across diverse agent ecosystems, cua_driver_acceleration provides a zero-dependency NPX installer (bin/cli.js) that auto-detects active agent harnesses and deploys the accelerated skill definitions without requiring git cloning:

```
# 1. Instant installation across all active agent harnesses:
npx github:pratikforge/cua_driver_acceleration
# 2. Target specific harness (antigravity, claude, codex, opencode, cursor, windsurf, roo):
npx github:pratikforge/cua_driver_acceleration --target antigravity
# 3. Read-only environment audit (checks cua-driver & agent-desktop availability):
npx github:pratikforge/cua_driver_acceleration --check
```

Supported Agent Harness Compatibility Matrix

Harness / Agent	Global Skills Location	Workspace Target	Status
Google Antigravity	~/.gemini/config/skills/	Project .gemini/	**Verified**
Claude Code	~/.claude/skills/	Project .claude/skills/	**Verified**
OpenAI Codex	~/.codex/skills/ (or $CODEX_HOME)	Project .codex/skills/	**Verified**
Cursor	N/A (Workspace root)	.cursor/skills/ or .cursorrules	**Verified**
OpenCode	~/.config/opencode/skills/	Project .opencode/skills/	Supported
Windsurf (Codeium)	~/.codeium/windsurf/skills/	Project .windsurf/skills/	Supported
Roo Code / Cline	N/A (Workspace root)	Project .roo/skills/	Supported
Universal Agent Standard	~/.agents/skills/	Project .agents/skills/	Supported

10. Operational Recommendations for Agent Architects

Rule 1: Ban Throwaway Script Synthesis. Agents should avoid generating ad-hoc Python files for single clicks or keystrokes when a direct CLI call exists. Standardize on a resident CLI daemon or persistent stdin pipe with pre-warmed OS hooks.

Rule 2: Prefer Semantic Accessibility over Pixel Estimation. Treat the desktop as a DOM—query accessibility trees and dispatch to element tokens rather than guessing coordinates.

Rule 3: Exploit OS Deep Links Before Touching the Mouse. Do not simulate manual Start menu navigation. Trigger direct URI schemes (ms-settings: on Windows; x-apple.systempreferences: as a macOS reference).

Rule 4: Batch Predictable Operations into Single Turns. Replace the reactive click -> screenshot -> think loop with deterministic sequential dispatch.

11. Conclusion, Licensing & Citation

By systematically replacing vision screenshots, Python script overhead, and multi-turn loops with accessibility DOM snapshots, stdin pipes, and URI deep links, cua_driver_acceleration enables sub-200ms per-action latency and high-reliability desktop workflows. The project is open-source under the MIT License at **https://github.com/pratikforge/cua_driver_acceleration**.

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@misc{yadav2026cua_acceleration,
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